

Software Requirements Specification (SRS)

Project: AI Mock Interview Platform

1. Introduction

Purpose: AI-powered mock interview platform using MERN and Groq API for technical and HR interview practice. Users can create interviews, answer AI-generated questions, receive evaluations, and track progress.

2. Technology Stack

Frontend: React (Vite), Tailwind CSS, React Router, Axios.

Backend: Node.js, Express.js.

Database: MongoDB Atlas.

AI: Groq API.

Authentication: JWT + bcrypt.

Deployment: Vercel (Frontend), Render (Backend), GitHub, MongoDB Atlas.

3. Functional Requirements

Authentication, Dashboard, Create Interview, AI Interview Flow, AI Evaluation, Final Report, Interview History, Profile Management.

4. Non-Functional Requirements

Performance (<3s typical response), Security (JWT, HTTPS, bcrypt, Helmet, CORS, Rate Limiting), Scalability, Availability.

5. Database Design

Users, Interviews, and Questions collections with references and timestamps.

6. REST APIs

/api/auth/register, /login, /profile; /api/interview/create, /start, /answer, /history, /:id.

7. UI Pages

Landing, Login, Register, Dashboard, Create Interview, Interview Screen, Feedback, Report, History, Profile.

8. AI Prompt Strategy

Question generation prompt based on role, experience, and difficulty. Evaluation prompt returns structured JSON with score, strengths, weaknesses, and ideal answer.

9. Future Enhancements

Voice interviews, Resume parsing, Coding interviews, Company-specific interviews, Analytics, PDF export, Payments, Admin Dashboard.

10. Milestones

Setup, Authentication, Dashboard, Interview Flow, Groq Integration, Reports, Testing, Deployment.